

Section B: Answer **ALL** questions. Parts marked with * involve knowledge of the extension component. Write your answers in the spaces provided.

1. The following experimental items are provided for estimating the specific heat capacity of bronze c_b :

- a bronze sphere of mass 0.80 kg hung with a thread at room temperature T_0
- a polystyrene cup containing 0.50 kg of water at room temperature T_0
- a water bath maintained at 80 °C
- a thermometer
- a stirrer
- a towel

(a) Describe the procedures of the experiment and state **TWO** experimental precautions to be taken. Write down an equation for finding c_b .

Given: specific heat capacity of water = 4200 J kg⁻¹ °C⁻¹

(6 marks)

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(b) The value of c_b found in the experiment in (a) is lower than the actual value. Explain. (2 marks)

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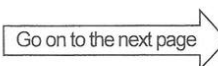
Book 1 section 2.3: DSE long questions

2016 Q1 (page 3)

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2016-DSE-PHY 1B-3

3



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1. In a restaurant, 'wontons in soup' is prepared by putting 5 pieces of cooked wonton at 4°C into a bowl with 0.60 kg of soup at temperature 96°C .

Given: average mass of each piece of wonton = 0.02 kg
 specific heat capacity of wonton = $3300\text{ J kg}^{-1}\text{ }^{\circ}\text{C}^{-1}$
 specific heat capacity of soup = $4200\text{ J kg}^{-1}\text{ }^{\circ}\text{C}^{-1}$

- (a) Find the final temperature of the mixture. Assume that the heat capacity of the bowl and the heat loss to the surroundings are negligible. (2 marks)

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- (b) The soup in (a) is taken from a metallic container of heat capacity $2000\text{ J }^{\circ}\text{C}^{-1}$ containing 16 kg of soup maintained at 96°C by an immersion heater.

- (i) Why does that energy have to be supplied by the heater to keep the soup at 96°C ? (1 mark)

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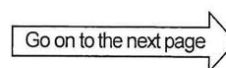
- (ii) A student used the following method to find the heater's operating power P : remove the heater from the container and record the temperature of the 16 kg of soup after 10 minutes. It is found that the temperature has dropped 9°C . Estimate P . (3 marks)

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- (iii) If the student repeats the measurement after another 10 minutes, would the corresponding temperature drop be larger than, equal to or smaller than 9°C ? Explain. (2 marks)

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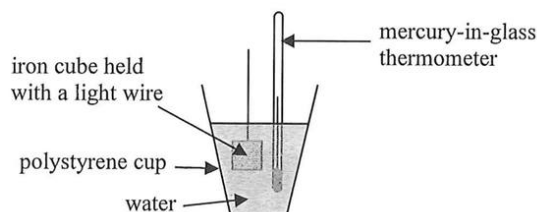
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1. An iron cube of mass 200 g is heated to a few hundred degree Celsius (say at $T^{\circ}\text{C}$). It is then quickly transferred into a polystyrene cup containing 600 g of water. The cube is totally immersed in the water as shown in Figure 1.1. The temperature of the heated iron cube can be estimated in this experiment.

Figure 1.1



Assume that no heat is lost to the surroundings and neglect the heat capacity of the polystyrene cup.

- (a) The temperature of water recorded by the thermometer rises from 25°C to 33°C . Estimate T .
Given: The specific heat capacities of iron and water are $450 \text{ J kg}^{-1} ^{\circ}\text{C}^{-1}$ and $4200 \text{ J kg}^{-1} ^{\circ}\text{C}^{-1}$ respectively. (3 marks)

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- (b) What would happen to the water when this iron cube of high temperature just touches the water surface? (1 mark)

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- (c) Explain why the actual temperature of the iron cube should be higher than that estimated in (a). (2 marks)

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- (d) (i) A mercury-in-glass thermometer is not suitable for measuring the temperature of the heated iron cube **directly**. Why? (1 mark)

- (ii) Suggest a kind of thermometer that can directly measure the temperature of this heated cube. (1 mark)

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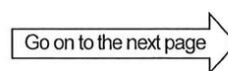
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Book 1 section 2.3: DSE long questions

2023 Q1 (page 3)

2023-DSE-PHY 1B-3

3



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1. An empty glass cup is placed on a table under room temperature of 23°C . 400 g of water at 100°C is poured into the cup. After a moment, the temperature of water in the cup is found to be 93°C .

(a) State a cause for the drop of water temperature other than heat loss to the surrounding air. (1 mark)

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(b) Estimate the heat capacity H , in $\text{J }^\circ\text{C}^{-1}$, of the glass cup. (2 marks)
Given: specific heat capacity of water = $4200 \text{ J kg}^{-1} ^\circ\text{C}^{-1}$

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(c) (i) What is the kind of electromagnetic waves, other than visible light, mainly emitted from this cup of hot water? (1 mark)

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(ii) Name a device in daily life which makes use of this kind of electromagnetic waves. (1 mark)

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